

HEALTH, SAFETY & ENVIRONMENT (HSE) – ME-407

HITEC UNIVERSITY, TAXILA

INSTRUCTOR'S INFORMATION

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INTRODUCTION

- ▶ Learning practical aspects of protecting the environment and maintaining health and safety at occupation.
- ▶ Learning principles that organizations and individuals must do to make sure that their activities do not cause harm to anyone.



COURSE OUTLINE

- ▶ Instructing students in the fundamentals of health, safety and environment.
- ▶ Building foundation for the study of safety, health and environmental issues.
- ▶ Providing guidelines to implement these concepts in professional organizations.

COURSE OUTLINE (contd.)

▶ Quizzes	05%
▶ Assignments	10%
▶ Project / Presentation	10%
▶ Sessionals / Midterm Exam	30%
▶ Final Exam	45%

COURSE OUTLINE (contd.)

- ▶ Introduction to Health & Safety
- ▶ Managing Risks
- ▶ Risk Management Techniques & Practices
- ▶ Practical Health & Safety Management
- ▶ Occupational Health & Safety Hazards
- ▶ Environmental Management

HEALTH, SAFETY & ENVIRONMENT

- ▶ **HEALTH**

Developing safe and environmentally-friendly processes to prevent / reduce the risk of harm to people.

- ▶ **SAFETY**

Creating organized procedures for identifying workplace hazards and reducing accidents.

- ▶ **ENVIRONMENT**

Creating a systematic approach to complying with environmental regulations.

HEALTH, SAFETY & ENVIRONMENT (contd.)

To minimize or eliminate hazards in our work / systems / projects.



SAFETY: *Importance*

► Moral



► Legal



► Financial



SAFETY: *Importance*

- ▶ Over 350,000 work-related fatal accidents each year.
- ▶ 2.75 million work-related fatalities each year.
- ▶ 2.4 million fatalities from occupational diseases.
- ▶ 270 million accidents and 160 million diseases a year due to work.
- ▶ 4% of global GDP is lost.

(Statistics from International Labor Organization's SafeWork Program)

SAFETY: *Types*

► Occupational Safety

► Public Safety



SAFETY: *Techniques*

- ▶ Safety Analysis
- ▶ Safety Reviews
- ▶ Risk Assessment
- ▶ Formal Inspections
- ▶ Hazard & Operability Study (HAZOP)
- ▶ Layers of Protection Analysis (LOPA)

SAFETY: *Designing It*

- ▶ Designing & building Safety in a system / project can be done by:
 - ▶ Designing out hazards
 - ▶ Creating Margin of Error
 - ▶ Building in Fail Safes
 - ▶ Safety Procedures



HAZARDS

- ▶ A situation that has the potential to cause injury or adverse health effects for people and damage to the plant and/or equipment.



OCCUPATIONAL HAZARDS

- ▶ Major Accidents in Workplace:

- ▶ Falls & Falling Objects
- ▶ Chemical Exposure
- ▶ Fire Hazards
- ▶ Electrical Hazards
- ▶ Repetitive Motion Injury

- ▶ Workplace Hazards:

1. Safety
2. Biological
3. Physical
4. Ergonomic
5. Chemical
6. Work Organization
7. Environmental

International Standards

- ▶ **ISO 45001** (Occupational Health & Safety)
- ▶ **ISO 12100** (Safety of Machinery)
- ▶ **ISO 14001** (Environmental Management)
- ▶ **ISO 9001** (Quality Management)

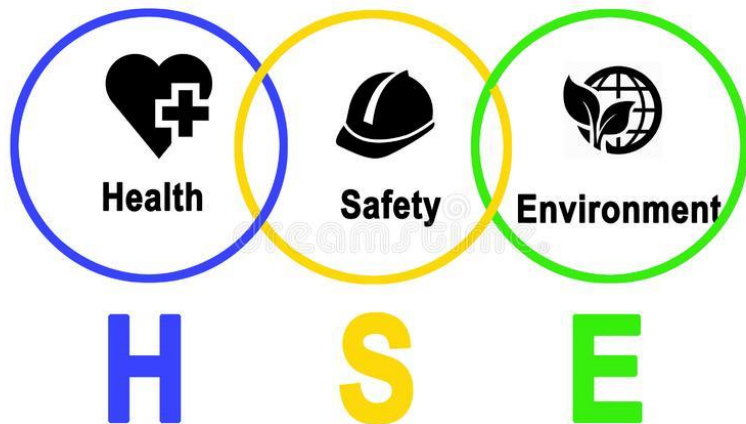
International Organizations

- ▶ **OSHA** (Occupational Safety & Health Administration)
- ▶ **ISO** (International Organization for Standardization)
- ▶ **ILO** (International Labor Organization)

- ▶ **IOSH** (Institute of Occupational Health & Safety)
- ▶ **NEBOSH** (National Examination Board in Occupational Safety & Health)

HSE: *Summary*

- ▶ Protecting people from environment
- ▶ Protecting environment from people
- ▶ Improving overall quality of workplace



CAUTION

**DO NOT OPERATE
THIS MACHINE WITHOUT
GUARDS IN PLACE**

CONCLUSION



THANK YOU FOR
YOUR LISTENING

DO YOU HAVE
ANY QUESTIONS?